

Interior-Point Methods for Linear Optimization

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1 Linear Optimization with a Logarithmic Barrier Function

Interior-point methods (also known as IPMs) for linear optimization are optimization methods based on the logarithmic barrier approach to solving a linear optimization problem. These methods date back to the work of Fiacco and McCormick in their 1967 book *Sequential Unconstrained Minimization Techniques*. At that time the methods were not believed then to be either practically or theoretically interesting. Interest in logarithmic barrier methods were re-born as a consequence of Karmarkar's revolutionary work on interior-point methods in 1984 [Jorge, need reference here], and have been the subject of an enormous amount of theoretical and applied research ever since. In this chapter we present three basic IPMs for linear optimization. The first method we present is based on Newton's method applied to the primal logarithmic barrier optimization problem. The second method is based Newton's method applied to primal-dual equations of the logarithmic barrier problem. The third method is a practical version of the second method, that is very close to what is used in practice in modern IPM software.

Consider the linear optimization problem in standard form:

$$\begin{aligned} P : \quad & \text{minimize}_x \quad c^T x \\ & \text{s.t.} \quad Ax = b \\ & \quad \quad x \geq 0 , \end{aligned} \tag{1}$$

where x is a vector of n variables, whose standard linear optimization dual problem is:

$$\begin{aligned} D : \quad & \text{maximize}_{\pi, s} \quad b^T \pi \\ & \text{s.t.} \quad A^T \pi + s = c \\ & \quad \quad s \geq 0 . \end{aligned} \tag{2}$$

Given a feasible solution x of P and a feasible solution (π, s) of D , the

duality gap is simply:

$$c^T x - b^T \pi = x^T s \geq 0 .$$

We introduce the following notation which will be very convenient for working with the mathematics of IPMs. Suppose that $x > 0$. Define the matrix X to be the $n \times n$ diagonal matrix whose diagonal entries are precisely the components of x . Then X looks like:

$$X = \begin{pmatrix} x_1 & 0 & \dots & 0 \\ 0 & x_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & x_n \end{pmatrix} .$$

Notice that X is positive definite, and so is X^2 , which looks like:

$$X^2 = \begin{pmatrix} (x_1)^2 & 0 & \dots & 0 \\ 0 & (x_2)^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & (x_n)^2 \end{pmatrix} .$$

Similarly, the matrices X^{-1} and X^{-2} look like:

$$X^{-1} = \begin{pmatrix} (1/x_1) & 0 & \dots & 0 \\ 0 & (1/x_2) & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & (1/x_n) \end{pmatrix}$$

and

$$X^{-2} = \begin{pmatrix} 1/(x_1)^2 & 0 & \dots & 0 \\ 0 & 1/(x_2)^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1/(x_n)^2 \end{pmatrix} .$$

Introducing the now-standard notation $e = (1, 1, 1, \dots, 1)^T$, we also have:

$$Xe = x \quad \text{and} \quad X^{-1}e = \begin{pmatrix} 1/x_1 \\ \vdots \\ 1/x_n \end{pmatrix} .$$

Let us append a logarithmic barrier term to the objective function of P , parameterized with a positive scalar θ , whereby our problem becomes:

$$\begin{aligned} P(\theta) : \quad & \text{minimize}_x \quad c^T x - \theta \sum_{j=1}^n \ln(x_j) \\ & \text{s.t.} \quad Ax = b \\ & \quad \quad x > 0 . \end{aligned}$$

One can think of the above problem as seeking to minimize the original objective function $c^T x$ plus the penalizing term

$$-\theta \sum_{j=1}^n \ln(x_j)$$

that repels solutions from the boundary inequalities $x_j \geq 0$, $j = 1, \dots, n$, where the penalty will be large or small to the extent that $\theta > 0$ is large or small.

The gradient of the objective function of $P(\theta)$ is simply $c - \theta X^{-1}e$, and the Karush-Kuhn-Tucker conditions for $P(\theta)$ therefore are:

$$\begin{cases} Ax = b, \ x > 0 \\ c - \theta X^{-1}e = A^T \pi . \end{cases} \quad (3)$$

If we define $s = \theta X^{-1}e$, then

$$\frac{1}{\theta} Xs = e ,$$

or equivalently

$$\frac{1}{\theta} X S e = e ,$$

and we can rewrite the Karush-Kuhn-Tucker conditions as:

$$\begin{cases} Ax = b, \ x > 0 \\ A^T \pi + s = c \\ \frac{1}{\theta} X S e - e = 0 . \end{cases} \quad (4)$$

From the equations of (4) it follows that if (x, π, s) is a solution of (4), then x is feasible for P , (π, s) is feasible for D , and the resulting duality gap is:

$$c^T x - b^T \pi = x^T s = e^T X S e = \theta e^T e = \theta n . \quad (5)$$

This suggests that we try solving $P(\theta)$ for a variety of values of θ as $\theta \rightarrow 0$.

For a given value of $\theta > 0$, let $(x(\theta), \pi(\theta), s(\theta))$ be the solution of (4). Then $x(\theta)$ is the optimal solution of $P(\theta)$. As θ varies over all positive values, the set

$$\{x(\theta) : \theta > 0\}$$

is called the *primal central path* (or the *primal central trajectory*) of the linear optimization problem (1). It is also common to refer to the the set

$$\{(x(\theta), \pi(\theta), s(\theta)) : \theta > 0\}$$

as the *primal-dual central path* (or the *primal-dual central trajectory*) of the duality-paired linear optimization problems (1)-(2). Furthermore, we will call the equation system (4) the *central path equations* or the *central trajectory equations*. Motivated by the observation from (5) that $(x(\theta))^T s(\theta) = n\theta$, interior-point methods (IPMs) seek to compute points along the central path for a sequence of values of θ as $\theta \rightarrow 0$.

However, we cannot usually solve (4) exactly, because the third equation group is not linear in the variables. We will instead work with an approximate solution of the Karush-Kuhn-Tucker conditions (4). We define (x, π, s) to be a “ β -approximate solution of $P(\theta)$ ” if (x, π, s) satisfies:

$$\begin{cases} Ax = b, \ x > 0 \\ A^T \pi + s = c \\ \|\frac{1}{\theta} Xs - e\| \leq \beta . \end{cases} \quad (6)$$

Here the norm $\|\cdot\|$ is the Euclidean norm. If (x, π, s) is a β -approximate solution of $P(\theta)$, then it will be useful to think of (x, π, s) as being “near” the central path of the linear optimization problem, where now nearness is not necessarily the physical distance, but rather is in the sense that the central path equation system (4) is nearly satisfied. Let us examine some properties of a β -approximate solution of $P(\theta)$.

Lemma 1.1 *If $(\bar{x}, \bar{\pi}, \bar{s})$ is a β -approximate solution of $P(\theta)$ and $\beta < 1$, then $\bar{x}$ is feasible for P , $(\bar{\pi}, \bar{s})$ is feasible for D , and the duality gap satisfies:*

$$n\theta(1 - \beta) \leq c^T \bar{x} - b^T \bar{\pi} = \bar{x}^T \bar{s} \leq n\theta(1 + \beta) . \quad (7)$$

Proof: Primal feasibility is obvious. To prove dual feasibility, we need to show that $\bar{s} \geq 0$. To see this, note that the third equation system of (6) implies that

$$-\beta \leq \frac{\bar{x}_j \bar{s}_j}{\theta} - 1 \leq \beta$$

which we can rearrange as:

$$\theta(1 - \beta) \leq \bar{x}_j \bar{s}_j \leq \theta(1 + \beta) . \quad (8)$$

Therefore $\bar{x}_j \bar{s}_j \geq (1 - \beta)\theta > 0$, which implies $\bar{x}_j \bar{s}_j > 0$, and so $\bar{s}_j > 0$. From (8) we have

$$n\theta(1 - \beta) = \sum_{j=1}^n \theta(1 - \beta) \leq \sum_{j=1}^n \bar{x}_j \bar{s}_j = \bar{x}^T \bar{s} \leq \sum_{j=1}^n \theta(1 + \beta) = n\theta(1 + \beta) .$$

■

As suggested by Lemma 1.1, we are motivated to consider a method for solving linear optimization that will compute a sequence of β -approximate solutions (x^k, π^k, s^k) of $P(\theta^k)$ for a decreasing sequence of values of $\theta^k \rightarrow 0$. We call such a method a “path-following” method. Algorithm 1 presents such a prototypical method.

Algorithm 1 Prototypical Central Path-Following Method for Linear Optimization

Step 0. Initialization. Data is $(x^0, \pi^0, s^0, \theta^0)$. $k = 0$. Assume that (x^0, π^0, s^0) is a β -approximate solution of $P(\theta^0)$ for some given value of β .

Step 1. Set current values. $(\bar{x}, \bar{\pi}, \bar{s}) \leftarrow (x^k, \pi^k, s^k)$, $\tilde{\theta} \leftarrow \theta^k$.

Step 2. Shrink θ . Set $\bar{\theta} \leftarrow \alpha \tilde{\theta}$ for some $\alpha \in (0, 1)$.

Step 3. Compute direction. Compute the direction $(\Delta x, \Delta \pi, \Delta s)$.

Step 4. Update All Values. $(x', \pi', s') \leftarrow (\bar{x}, \bar{\pi}, \bar{s}) + (\Delta x, \Delta \pi, \Delta s)$.

Step 5. Reset Counter and Continue. $(x^{k+1}, \pi^{k+1}, s^{k+1}) \leftarrow (x', \pi', s')$. $\theta^{k+1} \leftarrow \bar{\theta}$. $k \leftarrow k + 1$. Go to **Step 1**.

Algorithm 1 is initialized with values (x^0, π^0, s^0) that are a β -approximate solution of $P(\theta^0)$ for some given value of β and θ^0 . (Therefore the algorithm is assuming that x^0 and (π^0, s^0) are feasible primal and dual solutions, respectively, and that (somehow) the initial solution is also an approximate solution of $P(\theta^0)$. We will see later how this assumption can be relaxed in theory, as well as in practice.) In Step (2.) of the algorithm, the value of θ is shrunk from $\theta = \tilde{\theta} = \theta^k$ to $\theta = \bar{\theta} = \alpha \theta^k = \theta^{k+1}$,

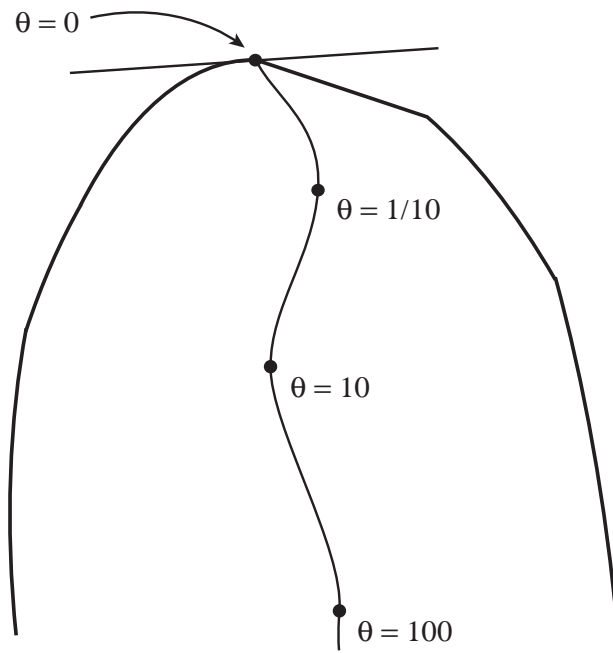


Figure 1: A conceptual picture the prototypical central path-following method described in Algorithm 1.

where the shrinkage factor is α and $0 < \alpha < 1$. Towards the goal of computing the next iterate $(x^{k+1}, \pi^{k+1}, s^{k+1})$, the algorithm in Step (3.) computes a direction $(\Delta x, \Delta \pi, \Delta s)$, and the next iterate values are set in Step (5.) as $(x^{k+1}, \pi^{k+1}, s^{k+1}) \leftarrow (x^k, \pi^k, s^k) + (\Delta x, \Delta \pi, \Delta s)$. Here the goal is to determine the direction $(\Delta x, \Delta \pi, \Delta s)$ in such a way that the next iterate $(x^{k+1}, \pi^{k+1}, s^{k+1})$ will hopefully be a β -approximate solution of $P(\bar{\theta}^{k+1})$. Figure 1 shows a conceptual picture of the prototypical path-following method described in Algorithm 1.

We will see that different approaches to the problem will lead to different methods for computing the direction $(\Delta x, \Delta \pi, \Delta s)$. In Section 2 we present a method where the direction in Step (3.) is determined based on a Newton step for the primal logarithmic barrier optimization problem. In Section 3 we present a different method, where the direction in Step (3.) is determined as a linearization of the central path equations (4). Last of all, in Section 4, we present a more practical version of the method of Section 3, that is very close to what is used in practice in modern IPM software.

[Jorge, we need a different version of Figure 1.]

2 An Interior-Point Method based on Newton's Method for the Primal Logarithmic Barrier Optimization Problem

In this section we present a more specific version of the prototypical path-following algorithm (Algorithm 1) based on Newton's method for the primal logarithmic barrier optimization problem.

Let $\bar{x}$ be a feasible solution to the logarithmic barrier problem $P(\theta)$:

$$P(\theta) : \quad \text{minimize}_x \quad c^T x - \theta \sum_{j=1}^n \ln(x_j)$$

$$\begin{aligned} \text{s.t.} \quad & Ax = b \\ & x > 0 . \end{aligned}$$

Our immediate interest is to construct the Newton step for $P(\theta)$ at $\bar{x}$. Let us denote the objective function of $P(\theta)$ by $f_\theta(x)$, namely,

$$f_\theta(x) = c^T x - \theta \sum_{j=1}^n \ln(x_j) .$$

Then we can easily derive formulas for the gradient and Hessian of $P(\theta)$:

$$\nabla f_\theta(\bar{x}) = c - \theta \bar{X}^{-1} e ,$$

and

$$H_\theta(\bar{x}) = \theta \bar{X}^{-2} .$$

Recall that in general the Newton step is the solution of the second-order (quadratic) approximation of an optimization problem at the current point. In this context the optimization problem is $P(\theta)$ and the current point is $x = \bar{x}$. We can write down the second-order approximation of $P(\theta)$ at $x = \bar{x}$, which is:

$$\begin{aligned} \text{minimize}_x \quad & f_\theta(\bar{x}) + \nabla f_\theta(\bar{x})^T (x - \bar{x}) + \frac{1}{2} (x - \bar{x})^T H_\theta(\bar{x}) (x - \bar{x}) \\ \text{s.t.} \quad & A(x - \bar{x}) = 0 . \end{aligned}$$

Now let $\Delta x := x - \bar{x}$. Substituting the values for the gradient and the Hessian from above, we rewrite the above optimization problem as the following equivalent problem:

$$\begin{aligned}
& \text{minimize}_{\Delta x} \quad (c - \theta \bar{X}^{-1}e)^T \Delta x + \frac{1}{2}(\Delta x)^T (\theta \bar{X}^{-2})(\Delta x) \\
& \text{s.t.} \quad A \Delta x = 0 .
\end{aligned} \tag{9}$$

The solution Δx of (9) is, by definition, the Newton direction for $P(\theta)$ at $x = \bar{x}$.

Notice that (9) is a linearly constrained strictly convex optimization problem. Therefore the Karush-Kuhn-Tucker (KKT) conditions for (9) are necessary and sufficient. These KKT condition can be written as follows:

$$\begin{cases} c - \theta \bar{X}^{-1}e + \theta \bar{X}^{-2} \Delta x = A^T \pi' \\ A \Delta x = 0 . \end{cases} \tag{10}$$

We will call the equations (10) the *Newton equations*. Let Δx and π' denote the solution to the Newton equations. We can manipulate (10) to yield the following explicit formulas for Δx and π' :

$$\begin{cases} \Delta x = \bar{X}[I - \bar{X}A^T(A\bar{X}^2A^T)^{-1}A\bar{X}](e - \frac{1}{\theta}\bar{X}c) \\ \pi' = (A\bar{X}^2A^T)^{-1}A\bar{X}(\bar{X}c - \theta e) . \end{cases} \tag{11}$$

Suppose that $(\Delta x, \pi')$ is a solution of the Newton equations (10). We obtain the new value of the primal variable x by taking the Newton step, namely,

$$x' = \bar{x} + \Delta x .$$

We can produce new values of the dual variables (π, s) by setting the new value of π to be π' and by setting $s' := c - A^T \pi'$. Using (10), then, we have that

$$s' = c - A^T \pi' = \theta \bar{X}^{-1}e - \theta \bar{X}^{-2} \Delta x = \theta \bar{X}^{-1}(e - \bar{X}^{-1} \Delta x) . \tag{12}$$

We have the following very powerful convergence theorem which demonstrates the quadratic convergence of Newton's method for this problem, with an explicit guarantee of the range in which quadratic convergence takes place.

Theorem 2.1 (Explicit Quadratic Convergence of Newton's Method).

Suppose that $(\bar{x}, \bar{\pi}, \bar{s})$ is a β -approximate solution of $P(\theta)$ and $\beta < 1$. Let $(\Delta x, \pi')$ be the solution to the Newton equations (10), and let

$$x' = \bar{x} + \Delta x$$

and

$$s' = c - A^T \pi' = \theta \bar{X}^{-1}(e - \bar{X}^{-1} \Delta x) .$$

Then (x', π', s') is a β^2 -approximate solution of $P(\theta)$. ■

The proof of Theorem 2.1 is deferred to the end of this Section. Motivated by Theorem 2.1, we present in Algorithm 2 a specific version of the prototypical algorithm (Algorithm 1) where the directions are computed using the Newton equations (10).

Algorithm 2 Algorithm for Linear Optimization based on Newton's Method for the Primal Logarithmic Barrier Optimization Problem

Step 0. Initialization. Data is $(x^0, \pi^0, s^0, \theta^0)$. $k = 0$. Assume that (x^0, π^0, s^0) is a β -approximate solution of $P(\theta^0)$ for some given value of β that satisfies $\beta < 1$.

Step 1. Set current values. $(\bar{x}, \bar{\pi}, \bar{s}) \leftarrow (x^k, \pi^k, s^k)$, $\tilde{\theta} \leftarrow \theta^k$.

Step 2. Shrink θ . Set $\bar{\theta} \leftarrow \alpha \tilde{\theta}$ for some $\alpha \in (0, 1)$.

Step 3. Compute the Newton direction. Compute the Newton step Δx for $P(\bar{\theta})$ at $x = \bar{x}$ by solving the following system of equations for the solution $(\Delta x, \pi')$:

$$\begin{cases} c - \bar{\theta} \bar{X}^{-1} e + \bar{\theta} \bar{X}^{-2} \Delta x = A^T \pi' \\ A \Delta x = 0 \end{cases} \quad (13)$$

Step 4. Update Values.

$$\begin{cases} x' &= \bar{x} + \Delta x \\ s' &= c - A^T \pi' \end{cases} \quad (14)$$

Step 5. Reset Counter and Continue. $(x^{k+1}, \pi^{k+1}, s^{k+1}) \leftarrow (x', \pi', s')$. $\theta^{k+1} \leftarrow \bar{\theta}$. $k \leftarrow k + 1$. Go to **Step 1**.

In order to execute Algorithm 2, we need to specify the values of α and β that will be used. Intuition suggests that it will be desirable to set α to be small in order for the values of θ^k to shrink quickly. However, intuition also suggests that if α is set too small, then the current point will no longer lie in the range of quadratic convergence of Newton's method. For a given value of β , the following theorem shows how to determine α for which the current point will still be in the range of quadratic convergence for Newton's method.

Theorem 2.2 (Shrinkage Theorem for Algorithm 2.) *Suppose that $(\bar{x}, \bar{\pi}, \bar{s})$ is a β -approximate solution of $P(\tilde{\theta})$ and $\beta < 1$. Let*

$$\alpha = 1 - \frac{\sqrt{\beta} - \beta}{\sqrt{\beta} + \sqrt{n}}$$

and let $\bar{\theta} = \alpha\tilde{\theta}$. Then $(\bar{x}, \bar{\pi}, \bar{s})$ is a $\sqrt{\beta}$ -approximate solution of $P(\bar{\theta})$.

Proof: The triplet $(\bar{x}, \bar{\pi}, \bar{s})$ satisfies $A\bar{x} = b$, $\bar{x} > 0$, and $A^T\bar{\pi} + \bar{s} = c$, and so it remains to show that

$$\left\| \frac{1}{\bar{\theta}} \bar{X} \bar{s} - e \right\| \leq \sqrt{\beta} .$$

First notice that the expression for α in the statement of the theorem can be manipulated to yield:

$$\alpha = \frac{\beta + \sqrt{n}}{\sqrt{\beta} + \sqrt{n}} . \quad (15)$$

We then have:

$$\begin{aligned} \left\| \frac{1}{\bar{\theta}} \bar{X} \bar{s} - e \right\| &= \left\| \frac{1}{\alpha\tilde{\theta}} \bar{X} \bar{s} - e \right\| \\ &= \left\| \frac{1}{\alpha} \left(\frac{1}{\tilde{\theta}} \bar{X} \bar{s} - e \right) - \left(1 - \frac{1}{\alpha} \right) e \right\| \\ &\leq \left(\frac{1}{\alpha} \right) \left\| \frac{1}{\tilde{\theta}} \bar{X} \bar{s} - e \right\| + \left| \frac{1 - \alpha}{\alpha} \right| \|e\| \\ &\leq \frac{\beta}{\alpha} + \left(\frac{1 - \alpha}{\alpha} \right) \sqrt{n} = \frac{\beta + \sqrt{n}}{\alpha} - \sqrt{n} = \sqrt{\beta} + \sqrt{n} - \sqrt{n} = \sqrt{\beta} , \end{aligned}$$

where the second-to-last equality above uses (15). ■

The following theorem states that if α is set according to the formula in Theorem 2.2, then Algorithm 2 will stay “on track,” namely all iterations will be β -approximate solutions for their respective values of θ .

Theorem 2.3 (Convergence Theorem for Algorithm 2). *Let $\beta < 1$ and let*

$$\alpha = 1 - \frac{\sqrt{\beta} - \beta}{\sqrt{\beta} + \sqrt{n}}$$

be the scalars used in Algorithm 2, and (x^k, π^k, s^k) be the iterates of Algorithm 2. If (x^0, π^0, s^0) is a β -approximate solution of $P(\theta^0)$, then (x^k, π^k, s^k) is a β -approximate solution of $P(\theta^k)$ for all $k \geq 1$.

Proof: We have by hypothesis that (x^k, π^k, s^k) is a β -approximate solution of $P(\theta^k)$ for $k = 0$. By induction, suppose that (x^k, π^k, s^k) is a β -approximate solution of $P(\theta^k)$ for some $k \geq 0$. From Theorem 2.2, (x^k, π^k, s^k) is a $\sqrt{\beta}$ -approximate solution of $P(\theta^{k+1})$ where $\theta^{k+1} = \alpha\theta^k$.

From the Theorem 2.1, $(x^{k+1}, \pi^{k+1}, s^{k+1})$ is a $(\sqrt{\beta})^2 = \beta$ -approximate solution of $P(\theta^{k+1})$. Therefore, by induction, the theorem is true for all values of k . ■

We now present specific computational guarantees for Algorithm 2, which follow as a consequence of Theorem 2.3.

Theorem 2.4 (Computational Guarantees for Algorithm 2). *Let $\beta = \frac{1}{4}$ and set*

$$\alpha = 1 - \frac{\sqrt{\beta} - \beta}{\sqrt{\beta} + \sqrt{n}}.$$

Suppose that (x^0, π^0, s^0) is a $\beta = \frac{1}{4}$ -approximate solution of $P(\theta^0)$. In order to obtain primal and dual feasible solutions (x^k, π^k, s^k) with a duality gap of at most ε , one needs to run Algorithm 2 for at most

$$k = \left\lceil 6\sqrt{n} \ln \left(\frac{1.25 (x^0)^T s^0}{0.75 \varepsilon} \right) \right\rceil$$

iterations.

Proof: Let k be as defined above. Note that

$$\alpha = 1 - \frac{\sqrt{\beta} - \beta}{\sqrt{\beta} + \sqrt{n}} = 1 - \frac{\frac{1}{2} - \frac{1}{4}}{\left(\frac{1}{2} + \sqrt{n}\right)} = 1 - \frac{1}{2 + 4\sqrt{n}} \leq 1 - \frac{1}{6\sqrt{n}}.$$

Therefore

$$\theta^k \leq \left(1 - \frac{1}{6\sqrt{n}}\right)^k \theta^0 .$$

This implies that

$$\begin{aligned} c^T x^k - b^T \pi^k &= (x^k)^T s^k \\ &\leq \theta^k n(1 + \beta) \\ &\leq \left(1 - \frac{1}{6\sqrt{n}}\right)^k (1.25n) \theta^0 \\ &\leq \left(1 - \frac{1}{6\sqrt{n}}\right)^k (1.25n) \left(\frac{(x^0)^T s^0}{0.75n}\right) , \end{aligned}$$

where the first and the third inequalities follow from (7). Taking logarithms, we obtain

$$\begin{aligned} \ln(c^T x^k - b^T \pi^k) &\leq k \ln \left(1 - \frac{1}{6\sqrt{n}}\right) + \ln \left(\frac{1.25}{0.75} (x^0)^T s^0\right) \\ &\leq \frac{-k}{6\sqrt{n}} + \ln \left(\frac{1.25}{0.75} (x^0)^T s^0\right) \\ &\leq -\ln \left(\frac{1.25}{0.75} \frac{(x^0)^T s^0}{\varepsilon}\right) + \ln \left(\frac{1.25}{0.75} (x^0)^T s^0\right) \\ &= \ln(\varepsilon) . \end{aligned}$$

Therefore $c^T x^k - b^T \pi^k \leq \varepsilon$. ■

It is important to note that from a theoretical point of view, Theorem 2.4 is rather remarkable. It states that for any linear optimization problem, regardless of its description or its data, the number of iterations needed to obtain an ε -optimal solution depends only on the initial duality of gap the initial point (x^0, π^0, s^0) (which is assumed to be a β -approximate solution of $P(\theta^0)$ for some θ^0), the desired accuracy ε , and $\sqrt{n}$. The result is independent of the data description or any other features of the linear optimization problem.

However, from a practical viewpoint Algorithm 2 is problematic in many ways. Note that Algorithm 2 assumes that the starting point (x^0, π^0, s^0)

is a β -approximate solution of $P(\theta^0)$ for some θ^0 . This is a rather strong assumption, as it assumes that x^0 and (π^0, s^0) are already primal and dual feasible solutions, and that (x^0, π^0, s^0) is nearly a solution to the logarithmic barrier optimization problem $P(\theta^0)$. In practice it is very rare that this assumption would be satisfied. Nevertheless, as is shown in Section 5, one can convert any linear optimization to a related problem for which the assumption is automatically satisfied. We refer the interested reader to Section 5 for a description of how the conversion can be done.

Another problem with Algorithm 2 (and its computational guarantees in Theorem 2.4) concerns the computational effort associated with the iteration bound in Theorem 2.4. At each iteration of Algorithm 2 the Newton equation system must be solved in Step (3.), and closed form formulas for this system are presented in equation (11). Notice from (11) that one must construct and solve a linear equation system whose matrix is $(A\bar{X}A^T)$. Forming and solving with this matrix involves m^2n operations in the worst case. Furthermore, according to Theorem 2.4, the number of iterations of Algorithm 2 is bounded above by $6\sqrt{n} \ln \left(\frac{1.25}{0.75} \frac{(x^0)^T s^0}{\varepsilon} \right)$. If $m = 1,000$, $n = 10,000$, $(x^0)^T s^0 = 10^1$, and $\varepsilon = 10^{-6}$, then the bound on the number of iterations itself is 9,977, and the total number of operations would then be $9.977 \times 10^{13} = 9,977 \times m^2n$. Clearly this is not practical nor is it competitive with other methods for solving linear optimization problems. In Section 4 we will present a practical version of a logarithmic barrier method, that is very close to what is used in practice in modern IPM software.

We end this section with the proof of Theorem 2.1.

Proof of Theorem 2.1: The current values $(\bar{x}, \bar{\pi}, \bar{s})$ satisfy:

$$\begin{cases} A\bar{x} = b, \bar{x} > 0 \\ A^T \bar{\pi} + \bar{s} = c \\ \|\frac{1}{\bar{\theta}} \bar{X} \bar{S} e - e\| \leq \beta . \end{cases} \quad (16)$$

Furthermore the Newton direction Δx and multipliers π' satisfy:

$$\begin{cases} A\Delta x = 0 \\ A^T \pi' + s' = c \\ x' = \bar{x} + \Delta x = \bar{X}(e + \bar{X}^{-1}\Delta x) \\ s' = c - A^T \pi' = \theta \bar{X}^{-1}e - \theta \bar{X}^{-2}\Delta x = \theta \bar{X}^{-1}(e - \bar{X}^{-1}\Delta x) . \end{cases} \quad (17)$$

We will first show that $\|\bar{X}^{-1}\Delta x\| \leq \beta$. It turns out that this is the key fact from which all results will follow. Note first that

$$A^T \bar{\pi} + \bar{s} = c = A^T \pi' + s' = A^T \pi' + \theta \bar{X}^{-1}e - \theta \bar{X}^{-2}\Delta x .$$

Multiplying the above equality on the left by Δx yields:

$$(\Delta x)^T \bar{s} = \theta (\Delta x)^T \bar{X}^{-1}e - \theta (\Delta x)^T \bar{X}^{-2}\Delta x ,$$

and so

$$\begin{aligned} (\|\bar{X}^{-1}\Delta x\|)^2 &= (\Delta x)^T \bar{X}^{-2}\Delta x = (\Delta x)^T \bar{X}^{-1}e - \frac{1}{\theta}(\Delta x)^T \bar{s} \\ &= (\Delta x)^T \bar{X}^{-1} \left(e - \frac{1}{\theta} \bar{X} \bar{s} \right) \\ &\leq \|\bar{X}^{-1}\Delta x\| \left\| e - \frac{1}{\theta} \bar{X} \bar{s} \right\| \\ &\leq \|\bar{X}^{-1}\Delta x\| \beta . \end{aligned}$$

This then implies that $\|\bar{X}^{-1}\Delta x\| \leq \beta$.

It therefore follows that

$$x' = \bar{X}(e + \bar{X}^{-1}\Delta x) > 0 ,$$

and

$$s' = \theta \bar{X}^{-1}(e - \bar{X}^{-1}\Delta x) > 0 .$$

Also, $A^T \pi' + s' = c$. Finally, we have for all $j = 1, \dots, n$, that:

$$\left(e - \frac{1}{\theta} X' s' \right)_j = 1 - \frac{1}{\theta} \bar{x}_j \left(1 + \frac{\Delta x_j}{\bar{x}_j} \right) \theta \left(\frac{1}{\bar{x}_j} \right) \left(1 - \frac{\Delta x_j}{\bar{x}_j} \right) = \left(\frac{\Delta x_j}{\bar{x}_j} \right)^2 .$$

Therefore

$$\left\| e - \frac{1}{\theta} X' s' \right\| \leq \sum_{j=1}^n \left| \left(e - \frac{1}{\theta} X' s' \right)_j \right| = \sum_{j=1}^n \left(\frac{\Delta x_j}{\bar{x}_j} \right)^2 = (\Delta x)^T \bar{X}^{-2} \Delta x = (\| \bar{X}^{-1} \Delta x \|^2) \leq \beta^2 ,$$

whereby (x', π', s') is a β^2 -approximate solution of $P(\theta)$. ■

3 A Primal-Dual Path Following IPM for Solving a Linear Optimization Problem

In Section 2 we motivated and studied the properties of Algorithm 2, which is a path-following method. Given current iterate values (x^k, π^k, s^k) , Algorithm 2 computes the new point $(x^{k+1}, \pi^{k+1}, s^{k+1})$ by using Newton's method for the primal logarithmic barrier problem $P(\theta)$. In so doing it solves the Newton equations (10) yielding the solution $(\Delta x, \pi')$ and the new dual slack variables are then set to $s' := c - A^T \pi'$. Therefore the new iterate values are in fact set as:

$$(x^{k+1}, \pi^{k+1}, s^{k+1}) \leftarrow (x^k, \pi^k, s^k) + (\Delta x, \pi', c - A^T \pi') ,$$

where $(\Delta x, \pi')$ are the solution of the Newton equations (10) for the logarithmic barrier problem $P(\theta)$. In the various theoretical results in Section 2, and in particular in Theorem 2.4, it was shown that Algorithm 2 possesses very powerful theoretical computational guarantees.

Notice in Algorithm 2 that its treatment of primal and dual variables is not the same. This is aesthetically unpleasing, and in particular it is also at odds with the similar roles that the primal and dual variables take in the central path equation system (4). In this section we consider a primal-dual path following IPM for which the primal and dual variables are treated in the same way. The method is based on using Newton's method to directly work with the central path equation system (4). Let us now see how this is done.

Let $(\bar{x}, \bar{\pi}, \bar{s})$ be our current iterate, which we assume is primal and dual feasible and for which the inequalities are strict, namely:

$$A\bar{x} = b, \quad \bar{x} > 0, \quad A^T \bar{\pi} + \bar{s} = c, \quad \bar{s} > 0. \quad (18)$$

Let $\bar{\theta}$ be the current value of the barrier parameter θ . Introducing a direction $(\Delta x, \Delta \pi, \Delta s)$, the next iterate will be $(\bar{x}, \bar{\pi}, \bar{s}) + (\Delta x, \Delta \pi, \Delta s)$, and we would like this iterate to solve the central path equations (4), namely:

$$\begin{cases} A(\bar{x} + \Delta x) = b, \bar{x} + \Delta x > 0 \\ A^T(\bar{\pi} + \Delta \pi) + (\bar{s} + \Delta s) = c \\ \frac{1}{\bar{\theta}}(\bar{X} + \Delta X)(\bar{S} + \Delta S)e - e = 0 . \end{cases}$$

Keeping in mind that $(\bar{x}, \bar{\pi}, \bar{s})$ is primal-dual feasible and so satisfies (18), we can rearrange the above to be:

$$\begin{cases} A\Delta x = 0 \\ A^T\Delta \pi + \Delta s = 0 \\ \bar{S}\Delta x + \bar{X}\Delta s = \bar{\theta}e - \bar{X}\bar{S}e - \Delta X\Delta Se . \end{cases}$$

Notice that the only nonlinear term in the above system of equations in $(\Delta x, \Delta \pi, \Delta s)$ is the term “ $\Delta X\Delta Se$ ” in the last system. This nonlinear term is also called a “second-order” term. If we erase this term, thereby ignoring the second-order term, we obtain the following equation system, which we call the primal-dual Newton equation system:

$$\begin{cases} A\Delta x = 0 \\ A^T\Delta \pi + \Delta s = 0 \\ \bar{S}\Delta x + \bar{X}\Delta s = \bar{\theta}e - \bar{X}\bar{S}e . \end{cases} \tag{19}$$

The solution $(\Delta x, \Delta \pi, \Delta s)$ of the system (19) is called the *primal-dual Newton step*. We can manipulate these equations to yield the following formulas for their solution:

$$\begin{aligned}
\Delta x &\leftarrow \left[I - \bar{X} \bar{S}^{-1} A^T (A \bar{X} \bar{S}^{-1} A^T)^{-1} A \right] (-\bar{x} + \bar{\theta} \bar{S}^{-1} e) , \\
\Delta \pi &\leftarrow (A \bar{X} \bar{S}^{-1} A^T)^{-1} A (\bar{x} - \bar{\theta} \bar{S}^{-1} e) , \\
\Delta s &\leftarrow A^T (A \bar{X} \bar{S}^{-1} A^T)^{-1} A (-\bar{x} + \bar{\theta} \bar{S}^{-1} e) .
\end{aligned} \tag{20}$$

Notice, by the way, that the computational effort in these equations lies primarily in solving the single equation system:

$$(A \bar{X} \bar{S}^{-1} A^T) \Delta \pi = A (\bar{x} - \bar{\theta} \bar{S}^{-1} e) .$$

Once this system is solved, we can easily substitute:

$$\begin{aligned}
\Delta s &\leftarrow -A^T \Delta \pi \\
\Delta x &\leftarrow -\bar{x} + \bar{\theta} \bar{S}^{-1} e - \bar{S}^{-1} \bar{X} \Delta s .
\end{aligned} \tag{21}$$

Suppose that $(\Delta x, \Delta \pi, \Delta s)$ is the (unique) solution of the primal-dual Newton system (19). We obtain the new value of the variables (x, π, s) by taking the Newton step:

$$(x', \pi', s') = (\bar{x}, \bar{\pi}, \bar{s}) + (\Delta x, \Delta \pi, \Delta s) .$$

We have the following very powerful convergence theorem which demonstrates the quadratic convergence of the primal-dual Newton's method for the central path equations, with an explicit guarantee of the range in which quadratic convergence takes place.

Theorem 3.1 (Explicit Quadratic Convergence of Primal-Dual Newton's Method). *Suppose that $(\bar{x}, \bar{\pi}, \bar{s})$ is a β -approximate solution of $P(\bar{\theta})$ and $\beta < \frac{1}{2}$. Let $(\Delta x, \Delta \pi, \Delta s)$ be the solution to the primal-dual Newton equations (19), and let:*

$$(x', \pi', s') = (\bar{x}, \bar{\pi}, \bar{s}) + (\Delta x, \Delta \pi, \Delta s) .$$

Then (x', π', s') is a $\left(\frac{1+\beta}{(1-\beta)^2}\right) \beta^2$ -approximate solution of $P(\bar{\theta})$.

The proof of Theorem 3.1 is deferred to the end of this Section. Motivated by Theorem 3.1, we present in Algorithm 3 another version of the prototypical algorithm (Algorithm 1) where the directions are computed using the primal-dual Newton equations (19).

Algorithm 3 Primal-Dual IPM for Linear Optimization

Step 0. Initialization. Data is $(x^0, \pi^0, s^0, \theta^0)$. $k = 0$. Assume that (x^0, π^0, s^0) is a β -approximate solution of $P(\theta^0)$ for some known value of β that satisfies $\beta < \frac{1}{3}$.

Step 1. Set current values. $(\bar{x}, \bar{\pi}, \bar{s}) \leftarrow (x^k, \pi^k, s^k)$, $\tilde{\theta} \leftarrow \theta^k$.

Step 2. Shrink θ . Set $\bar{\theta} \leftarrow \alpha \tilde{\theta}$ for some $\alpha \in (0, 1)$.

Step 3. Compute the primal-dual Newton direction. Compute the Newton step $(\Delta x, \Delta \pi, \Delta s)$ for the equation system (19) at $(x, \pi, s) = (\bar{x}, \bar{\pi}, \bar{s})$ for $\theta = \bar{\theta}$, by solving the following system of equations in the variables $(\Delta x, \Delta \pi, \Delta s)$:

$$\begin{cases} A\Delta x = 0 \\ A^T \Delta \pi + \Delta s = 0 \\ \bar{S}\Delta x + \bar{X}\Delta s = \bar{\theta}e - \bar{X}\bar{S}e. \end{cases} \quad (22)$$

Step 4. Update All Values. $(x', \pi', s') \leftarrow (\bar{x}, \bar{\pi}, \bar{s}) + (\Delta x, \Delta \pi, \Delta s)$.

Step 5. Reset Counter and Continue. $(x^{k+1}, \pi^{k+1}, s^{k+1}) \leftarrow (x', \pi', s')$. $\theta^{k+1} \leftarrow \bar{\theta}$. $k \leftarrow k + 1$. Go to **Step 1**.

In order to execute Algorithm 3, we need to specify the values of α and β

that will be used. Herein we use the values:

$$\beta = \frac{3}{40} \quad \text{and} \quad \alpha = 1 - \frac{\frac{1}{8}}{\frac{1}{5} + \sqrt{n}} .$$

The following theorem shows that with these values of α and β the current iterate still is in the range of quadratic convergence after the θ has been decreased.

Theorem 3.2 (Shrinkage Theorem for Algorithm 3.) *Suppose that $(\bar{x}, \bar{\pi}, \bar{s})$ is a $\beta = \frac{3}{40}$ -approximate solution of $P(\theta)$. Let*

$$\alpha = 1 - \frac{\frac{1}{8}}{\frac{1}{5} + \sqrt{n}}$$

and let $\bar{\theta} = \alpha\tilde{\theta}$. Then $(\bar{x}, \bar{\pi}, \bar{s})$ is a $\beta = \frac{1}{5}$ -approximate solution of $P(\bar{\theta})$.

Proof: The triplet $(\bar{x}, \bar{\pi}, \bar{s})$ satisfies $A\bar{x} = b$, $\bar{x} > 0$, and $A^T\bar{\pi} + \bar{s} = c$, and so it remains to show that

$$\left\| \frac{1}{\bar{\theta}} \bar{X} \bar{s} - e \right\| \leq \frac{1}{5} .$$

We have:

$$\begin{aligned} \left\| \frac{1}{\bar{\theta}} \bar{X} \bar{s} - e \right\| &= \left\| \frac{1}{\alpha\tilde{\theta}} \bar{X} \bar{s} - e \right\| \\ &= \left\| \frac{1}{\alpha} \left(\frac{1}{\tilde{\theta}} \bar{X} \bar{s} - e \right) - \left(1 - \frac{1}{\alpha} \right) e \right\| \\ &\leq \left(\frac{1}{\alpha} \right) \left\| \frac{1}{\tilde{\theta}} \bar{X} \bar{s} - e \right\| + \left| \frac{1 - \alpha}{\alpha} \right| \|e\| \\ &\leq \frac{\frac{3}{40}}{\alpha} + \left(\frac{1 - \alpha}{\alpha} \right) \sqrt{n} = \frac{\frac{3}{40} + \sqrt{n}}{\alpha} - \sqrt{n} = \frac{1}{5} . \end{aligned}$$

■

The following theorem states that if α and β are specified as in Theorem 3.2, then Algorithm 3 will stay “on track,” namely all iterations will be β -approximate solutions for their respective values of θ .

Theorem 3.3 (Convergence Theorem for Algorithm 3). *Let $\beta = \frac{3}{40}$ and let*

$$\alpha = 1 - \frac{\frac{1}{8}}{\frac{1}{5} + \sqrt{n}}$$

be the scalars used in Algorithm 3, and (x^k, π^k, s^k) be the iterates of Algorithm 3. If (x^0, π^0, s^0) is a β -approximate solution of $P(\theta^0)$, then (x^k, π^k, s^k) is a β -approximate solution of $P(\theta^k)$ for all $k \geq 1$.

Proof: We have by hypothesis that (x^k, π^k, s^k) is a β -approximate solution of $P(\theta^k)$ for $k = 0$. By induction, suppose that (x^k, π^k, s^k) is a β -approximate solution of $P(\theta^k)$ for some $k \geq 0$. From Theorem 3.2, (x^k, π^k, s^k) is a $\frac{1}{5}$ -approximate solution of $P(\theta^{k+1})$ where $\theta^{k+1} = \alpha\theta^k$. From Theorem 3.1, $(x^{k+1}, \pi^{k+1}, s^{k+1})$ is a $\hat{\beta}$ -approximate solution of $P(\theta^{k+1})$ for

$$\hat{\beta} = \frac{1 + \frac{1}{5}}{(1 - \frac{1}{5})^2} \left(\frac{1}{5}\right)^2 = \frac{3}{40} = \beta.$$

Therefore, by induction, the theorem is true for all values of k . ■

We now present specific computational guarantees for Algorithm 3, which follow as a consequence of Theorem 3.3.

Theorem 3.4 (Computational Guarantees for Algorithm 3). *Suppose that (x^0, π^0, s^0) is a $\beta = \frac{3}{40}$ -approximate solution of $P(\theta^0)$. In order to obtain primal and dual feasible solutions (x^k, π^k, s^k) with a duality gap of at most ε , one needs to run Algorithm 3 for at most*

$$k = \left\lceil 10\sqrt{n} \ln \left(\frac{43}{37} \frac{(x^0)^T s^0}{\varepsilon} \right) \right\rceil$$

iterations.

Proof: Let k be as defined above. Note that

$$\alpha = 1 - \frac{\frac{1}{8}}{\frac{1}{5} + \sqrt{n}} = 1 - \frac{1}{\left(\frac{8}{5} + 8\sqrt{n}\right)} \leq 1 - \frac{1}{10\sqrt{n}}.$$

Therefore

$$\theta^k \leq \left(1 - \frac{1}{10\sqrt{n}}\right)^k \theta^0.$$

This implies that

$$\begin{aligned} c^T x^k - b^T \pi^k &= (x^k)^T s^k \\ &\leq \theta^k n(1 + \beta) \\ &\leq \left(1 - \frac{1}{10\sqrt{n}}\right)^k \left(\frac{43}{40}n\right) \theta^0 \\ &\leq \left(1 - \frac{1}{10\sqrt{n}}\right)^k \left(\frac{43}{40}n\right) \left(\frac{(x^0)^T s^0}{\frac{37}{40}n}\right), \end{aligned}$$

where the first and the third inequalities follow from (7). Taking logarithms, we obtain

$$\begin{aligned} \ln(c^T x^k - b^T \pi^k) &\leq k \ln\left(1 - \frac{1}{10\sqrt{n}}\right) + \ln\left(\frac{43}{37}(x^0)^T s^0\right) \\ &\leq \frac{-k}{10\sqrt{n}} + \ln\left(\frac{43}{37}(x^0)^T s^0\right) \\ &\leq -\ln\left(\frac{43}{37} \frac{(x^0)^T s^0}{\varepsilon}\right) + \ln\left(\frac{43}{37}(x^0)^T s^0\right) \\ &= \ln(\varepsilon). \end{aligned}$$

Therefore $c^T x^k - b^T \pi^k \leq \varepsilon$. ■

Similar to Theorem 2.4, Theorem 3.4 is rather remarkable. It states that for any linear optimization problem, regardless of its description or its data, the number of iterations of Algorithm 3 needed to obtain an ε -optimal solution depends only on the initial duality of gap the initial point (x^0, π^0, s^0) (which is assumed to be a β -approximate solution of $P(\theta^0)$ for some θ^0), the desired

accuracy ε , and $\sqrt{n}$. The result is independent of the data description or any other features of the linear optimization problem.

However, from a practical viewpoint Algorithm 3 is problematic in many ways. Note that Algorithm 3 assumes that the starting point (x^0, π^0, s^0) is a β -approximate solution of $P(\theta^0)$ for some θ^0 . This is a rather strong assumption, as it assumes that x^0 and (π^0, s^0) are already primal and dual feasible solutions, and that (x^0, π^0, s^0) is nearly a solution to the logarithmic barrier optimization problem $P(\theta^0)$. In practice it is very rare that this assumption would be satisfied. Nevertheless, as is shown in Section 5, one can convert any linear optimization to a related problem for which the assumption is automatically satisfied. We refer the interested reader to Section 5 for a description of how the conversion can be done.

Another problem with Algorithm 3 (and its computational guarantees in Theorem 3.4) concerns the computational effort associated with the iteration bound in Theorem 3.4. At each iteration of Algorithm 3 the Newton equation system must be solved in Step (3.), and closed form formulas for this system are presented in equation (20). Notice from (20) that one must construct and solve a linear equation system whose matrix is $(A\bar{X}\bar{S}^{-1}A^T)$. Forming and solving with this matrix involves m^2n operations in the worst case. Furthermore, according to Theorem 3.4, the number of iterations of Algorithm 3 is bounded above by $10\sqrt{n} \ln \left(\frac{43}{37} \frac{(x^0)^T s^0}{\varepsilon} \right)$. If $m = 1,000$, $n = 10,000$, $(x^0)^T s^0 = 10^1$, and $\varepsilon = 10^{-6}$, then the bound on the number of iterations itself is 16,268, and the total number of operations would then be $1.627 \times 10^{14} = 16,268 \times m^2n$. Clearly this is not practical nor is it competitive with other methods for solving linear optimization problems. In Section 4 we will present a practical version of a logarithmic barrier method, that is very close to what is used in practice in modern IPM software.

We end this section with the proof of Theorem 3.1.

Proof of Theorem 3.1: Our current point $(\bar{x}, \bar{\pi}, \bar{s})$ satisfies:

$$\begin{cases} A\bar{x} = b, \bar{x} > 0 \\ A^T \bar{\pi} + \bar{s} = c \\ \|\frac{1}{\bar{\theta}} \bar{X} \bar{S} e - e\| \leq \beta . \end{cases} \quad (23)$$

Furthermore the primal-dual Newton step $(\Delta x, \Delta \pi, \Delta s)$ satisfies:

$$\begin{cases} A\Delta x = 0 \\ A^T \Delta \pi + \Delta s = 0 \\ \bar{S} \Delta x + \bar{X} \Delta s = \bar{\theta} e - \bar{X} \bar{S} e . \end{cases} \quad (24)$$

Note from the first two equations of (24) that $\Delta x^T \Delta s = 0$. From the third equation of (23) we have

$$1 - \beta \leq \frac{\bar{s}_j \bar{x}_j}{\bar{\theta}} \leq 1 + \beta, \quad j = 1, \dots, n, \quad (25)$$

which implies:

$$\bar{x}_j \geq \frac{(1 - \beta) \bar{\theta}}{\bar{s}_j} \text{ and } \bar{s}_j \geq \frac{(1 - \beta) \bar{\theta}}{\bar{x}_j}, \quad j = 1, \dots, n. \quad (26)$$

As a result of this we obtain:

$$\begin{aligned} \bar{\theta}(1 - \beta) \|\bar{X}^{-1} \Delta x\|^2 &= \bar{\theta}(1 - \beta) \Delta x^T \bar{X}^{-1} \bar{X}^{-1} \Delta x \\ &\leq \Delta x^T \bar{X}^{-1} \bar{S} \Delta x \\ &= \Delta x^T \bar{X}^{-1} (\bar{\theta} e - \bar{X} \bar{S} e - \bar{X} \Delta s) \\ &= \Delta x^T \bar{X}^{-1} (\bar{\theta} e - \bar{X} \bar{S} e) \\ &\leq \|\bar{X}^{-1} \Delta x\| \|\bar{\theta} e - \bar{X} \bar{S} e\| \\ &\leq \|\bar{X}^{-1} \Delta x\| \beta \bar{\theta} . \end{aligned}$$

From this it follows that

$$\|\bar{X}^{-1}\Delta x\| \leq \frac{\beta}{1-\beta} < 1 .$$

Therefore

$$x' = \bar{x} + \Delta x = \bar{X}(e + \bar{X}^{-1}\Delta x) > 0 .$$

We have the exact same chain of inequalities for the dual slack variables:

$$\begin{aligned} \bar{\theta}(1-\beta)\|\bar{S}^{-1}\Delta s\|^2 &= \bar{\theta}(1-\beta)\Delta s^T \bar{S}^{-1} \bar{S}^{-1} \Delta s \\ &\leq \Delta s^T \bar{S}^{-1} \bar{X} \Delta s \\ &= \Delta s^T \bar{S}^{-1} (\bar{\theta}e - \bar{X}\bar{S}e - \bar{S}\Delta x) \\ &= \Delta s^T \bar{S}^{-1} (\bar{\theta}e - \bar{X}\bar{S}e) \\ &\leq \|\bar{S}^{-1}\Delta s\| \|\bar{\theta}e - \bar{X}\bar{S}e\| \\ &\leq \|\bar{S}^{-1}\Delta s\| \beta \bar{\theta} . \end{aligned}$$

From this it follows that

$$\|\bar{S}^{-1}\Delta s\| \leq \frac{\beta}{1-\beta} < 1 .$$

Therefore

$$s' = \bar{s} + \Delta s = \bar{S}(e + \bar{S}^{-1}\Delta s) > 0 .$$

Next note from (24) that for $j = 1, \dots, n$ we have:

$$x'_j s'_j = (\bar{x}_j + \Delta x_j)(\bar{s}_j + \Delta s_j) = \bar{x}_j \bar{s}_j + \bar{x}_j \Delta s_j + \Delta x_j \bar{s}_j + \Delta x_j \Delta s_j = \bar{\theta} + \Delta x_j \Delta s_j .$$

Therefore

$$\left(e - \frac{1}{\bar{\theta}} X' S' e \right)_j = -\frac{\Delta x_j \Delta s_j}{\bar{\theta}} .$$

From this we obtain:

$$\begin{aligned}
\|e - \frac{1}{\theta} X' S' e\| &\leq \|e - \frac{1}{\theta} X' S' e\|_1 \\
&= \sum_{j=1}^n \frac{|\Delta x_j \Delta s_j|}{\theta} \\
&= \sum_{j=1}^n \frac{|\Delta x_j|}{\bar{x}_j} \frac{|\Delta s_j|}{\bar{s}_j} \frac{\bar{x}_j \bar{s}_j}{\theta} \\
&\leq \sum_{j=1}^n \frac{|\Delta x_j|}{\bar{x}_j} \frac{|\Delta s_j|}{\bar{s}_j} (1 + \beta) \\
&\leq \|\bar{X}^{-1} \Delta x\| \|\bar{S}^{-1} \Delta s\| (1 + \beta) \\
&\leq \left(\frac{\beta}{1 - \beta} \right)^2 (1 + \beta) .
\end{aligned}$$

■

4 A Practical Primal-Dual Interior-Point Algorithm

As mentioned at the end of Section 3, the primal-dual IPM described in Algorithm 3 has two chief drawbacks from a practical point of view. First, the algorithm assumes that the initial values are primal and dual feasible (and are near the central path). And second, the fractional decrease parameter α is set very close to 1.0, and as a result the prescribed number of iterations is too large even for reasonably small values of the dimension n . Herein we describe a practical primal-dual interior-point algorithm that overcomes these two drawbacks. More specifically, the algorithm described in this Section does not assume that the current point is feasible, nor near the central path. Also, the fractional decrease parameter α is effectively set to $\alpha = \frac{1}{10}$ rather than the theoretically necessary (and conservative) value of

$$\alpha = 1 - \frac{\frac{1}{8}}{\frac{1}{5} + \sqrt{n}} .$$

The algorithm does not always take the full Newton step at each iteration, and in fact utilizes different step-sizes for the primal and the dual variables.

In the algorithm the current “point” is $(\bar{x}, \bar{\pi}, \bar{s})$ for which $\bar{x} > 0$ and $\bar{s} > 0$, but quite possibly $A\bar{x} \neq b$ and/or $A^T\bar{\pi} + \bar{s} \neq c$. We are given a value of the central path barrier parameter $\bar{\theta} > 0$. We want to compute a direction $(\Delta x, \Delta \pi, \Delta s)$ so that $(\bar{x} + \Delta x, \bar{\pi} + \Delta \pi, \bar{s} + \Delta s)$ approximately solves the central path equations (4). We set up the system:

$$\begin{cases} A(\bar{x} + \Delta x) = b \\ A^T(\bar{\pi} + \Delta \pi) + (\bar{s} + \Delta s) = c \\ (\bar{X} + \Delta X)(\bar{S} + \Delta S)e = \bar{\theta}e . \end{cases} \quad (27)$$

We linearize this system of equations and rearrange the terms to obtain the Newton equations for the current point $(\bar{x}, \bar{\pi}, \bar{s})$:

$$\begin{cases} A\Delta x = b - A\bar{x} =: r_1 \\ A^T\Delta \pi + \Delta s = c - A^T\bar{\pi} - \bar{s} =: r_2 \\ \bar{S}\Delta x + \bar{X}\Delta s = \bar{\theta}e - \bar{X}\bar{S}e =: r_3 \end{cases} \quad (28)$$

We refer to the solution $(\Delta x, \Delta \pi, \Delta s)$ to (28) as the primal-dual Newton direction at the point $(\bar{x}, \bar{\pi}, \bar{s})$. This differs from the solution of the system (19) presented earlier only in that (19) assumed that $r_1 = 0$ and $r_2 = 0$.

Given our current point $(\bar{x}, \bar{\pi}, \bar{s})$ and a given value of $\bar{\theta}$, we compute the Newton direction $(\Delta x, \Delta \pi, \Delta s)$. If we were to take the full Newton step in both the primal variables x and the dual variables (π, s) , our new iterate would then be $(\bar{x} + \Delta x, \bar{\pi} + \Delta \pi, \bar{s} + \Delta s) = (\bar{x}, \bar{\pi}, \bar{s}) + (\Delta x, \Delta \pi, \Delta s)$. However, there is no guarantee that such an assignment would result in feasible values of x and (π, s) . In particular there is no guarantee that the full Newton step will result in either $\bar{x} + \Delta x > 0$ or $\bar{s} + \Delta s > 0$. Let us therefore consider determining step-sizes α_P for the primal update, and α_D for the dual update, to obtain new iterate values:

$$(\tilde{x}, \tilde{\pi}, \tilde{s}) \leftarrow (\bar{x} + \alpha_P \Delta x, \bar{\pi} + \alpha_D \Delta \pi, \bar{s} + \alpha_D \Delta s) .$$

In order to ensure that $\tilde{x} > 0$ and $\tilde{s} > 0$, we choose a value of r satisfying $0 < r < 1$ ($r = 0.99$ is a typical value used in practice), and determine α_P and α_D as follows:

$$\begin{aligned} \alpha_P &= \min \left\{ 1, r \min_{\Delta x_j < 0} \left\{ \frac{\bar{x}_j}{-\Delta x_j} \right\} \right\} \\ \alpha_D &= \min \left\{ 1, r \min_{\Delta s_j < 0} \left\{ \frac{\bar{s}_j}{-\Delta s_j} \right\} \right\} . \end{aligned}$$

These step-sizes ensure that the next iterate $(\tilde{x}, \tilde{\pi}, \tilde{s})$ satisfies $\tilde{x} > 0$ and $\tilde{s} > 0$.

4.1 Decreasing the Path Parameter θ

We also want to aggressively shrink θ at each iteration, in order to (hopefully) shrink the duality gap. The current iterate is $(\bar{x}, \bar{\pi}, \bar{s})$. Motivated by equation (5), if our current point $(\bar{x}, \bar{\pi}, \bar{s})$ was feasible and was on the central path for some $\tilde{\theta} > 0$, then from (5) it would have to be true that

$$\tilde{\theta} = \frac{\bar{x}^T \bar{s}}{n} .$$

If we simply define $\tilde{\theta}$ this way, it follows that:

$$\bar{x}^T \bar{s} = n \tilde{\theta} .$$

Having defined $\tilde{\theta}$, we then aggressively set

$$\bar{\theta} \leftarrow \left(\frac{1}{10} \right) \tilde{\theta} = \left(\frac{1}{10} \right) \left(\frac{\bar{x}^T \bar{s}}{n} \right) ,$$

where the fractional decrease $\alpha = \frac{1}{10}$ could be replaced by some other user-specified value. (Indeed, in practice it is common to use $\alpha = \frac{1}{10}$.)

4.2 The Stopping Criterion

We typically declare the linear optimization problem to be solved when it is “almost” primal feasible, “almost” dual feasible, and there is “almost” no duality gap. We set our tolerance ε to be a small positive number, typically $\varepsilon = 10^{-8}$, for example, and we stop when:

$$\max \{ \|A\bar{x} - b\|, \|A^T\bar{\pi} + \bar{s} - c\|, \bar{s}^T\bar{x} \} \leq \varepsilon .$$

4.3 The Complete Interior-Point Algorithm

The complete practical interior-point algorithm is presented in Algorithm 4.

The algorithm described in Algorithm 4 is almost exactly what is used in commercial interior-point method software. Computational experience with a huge number of linear optimization problems has demonstrated that a well-implemented interior-point algorithm, such as that of Algorithm 4, will solve almost any linear optimization problem in 25–35 iterations, regardless of the dimension of the problem. Although Algorithm 4 and other practical IPMs will solve almost any linear optimization in a very few iterations, there is no theoretical guarantee associated with Algorithm 4 or any other practical IPM. This is a paradox of theory and practice – the practical algorithm is not guaranteed mathematically to be efficient or even convergent, yet the theoretically efficient algorithms (such as Algorithm 2 and Algorithm 3) have their parameters α and β set in such way that they are inefficient in practice.

The theory of interior-point methods has been extended to cover virtually the entirety of convex nonlinear optimization problems. This will be shown later in the notes.

Algorithm 4 Practical Primal-Dual IPM for Linear Optimization

Step 0. Initialization. Given (x^0, π^0, s^0) satisfying $x^0 > 0$, $s^0 > 0$, and r satisfying $0 < r < 1$, $\varepsilon > 0$, and $\alpha \in (0, 1)$. Set $k \leftarrow 0$.

Step 1. Check Stopping Criterion. If:

$$\max \left\{ \|Ax^k - b\|, \|A^T \pi^k + s^k - c\|, (s^k)^T x^k \right\} \leq \varepsilon,$$

then STOP. Else proceed.

Step 2. Shrink θ . Set $\bar{\theta} \leftarrow (\alpha) \left(\frac{(x^k)^T (s^k)}{n} \right)$.

Step 3. Solve the Newton system. Solve the equations:

$$\begin{cases} A\Delta x = b - Ax^k =: r_1 \\ A^T \Delta \pi + \Delta s = c - A^T \pi^k - s^k =: r_2 \\ S^k \Delta x + X^k \Delta s = \bar{\theta} e - X^k S^k e =: r_3 \end{cases}$$

Step 4. Determine the Step-sizes. Compute:

$$\begin{aligned} \alpha_P &= \min \left\{ 1, r \min_{\Delta x_j < 0} \left\{ \frac{x_j^k}{-\Delta x_j} \right\} \right\} \\ \alpha_D &= \min \left\{ 1, r \min_{\Delta s_j < 0} \left\{ \frac{s_j^k}{-\Delta s_j} \right\} \right\}. \end{aligned}$$

Step 5. Update values.

$$(x^{k+1}, \pi^{k+1}, s^{k+1}) \leftarrow (x^k + \alpha_P \Delta x, \pi^k + \alpha_D \Delta \pi, s^k + \alpha_D \Delta s).$$

Re-set $k \leftarrow k + 1$ and return to **Step 2**.

5 Advanced Topic: How to Convert any Linear Optimization Problem to an Instance with an Initial Solution Near the Central Path

In Sections 2 and 3, we presented two algorithms for computing an ε -solution of a linear optimization problem (P) starting from a primal-dual feasible solution (x^0, π^0, s^0) that is near the central path, i.e., for which (x^0, π^0, s^0) is a β -approximate solution to the logarithmic barrier problem $P(\theta)$ for some given value of $\theta = \theta^0$. Note, however, that except in rare circumstances, we usually do not have available such a primal-dual feasible solution with this special property with which to start an algorithm. Here we show how, in theory, one can construct an augmented primal problem $(\tilde{P})$ for which such a starting primal-dual solution is readily available, and for which the optimal solutions of $(\tilde{P})$ and of (P) are the same.

Suppose that the original linear optimization problem P is given by:

$$\begin{aligned} P : \quad & \text{minimize} \quad c^T x \\ & \text{s.t.} \quad Ax = b \\ & \quad \quad x \geq 0 . \end{aligned}$$

For very large constants M and R , consider the following augmented problem $\tilde{P}$:

$$\tilde{P} : \quad \text{minimize} \quad c^T x + Mx_{n+1}$$

$$\text{s.t.} \quad Ax + \left(\frac{1}{R}b - Ae\right)x_{n+1} = \frac{1}{R}b$$

$$e^T x + x_{n+1} + x_{n+2} = n + 2$$

$$x \geq 0, \quad x_{n+1} \geq 0, \quad x_{n+2} \geq 0 .$$

The dual problem of $\tilde{P}$ is:

$$\tilde{D} : \quad \text{maximize} \quad \frac{1}{R}b^T \pi + (n+2)\pi_{m+1}$$

$$\pi, \pi_{m+1}, s, s_{n+1}, s_{n+2}$$

$$\text{s.t.} \quad A^T \pi + e\pi_{m+1} + s = c$$

$$\left(\frac{1}{R}b - Ae\right)^T \pi + \pi_{m+1} + s_{n+1} = M$$

$$\pi_{m+1} + s_{n+2} = 0$$

$$s, s_{n+1}, s_{n+2} \geq 0 .$$

First notice that as long as M and R are sufficiently large, then any basic optimal solution $(x^*, x_{n+1}^*, x_{n+2}^*)$ of $\tilde{P}$ will have $x_{n+1}^* = 0$ and $x_{n+2}^* > 0$, and so Rx^* will be an optimal solution of P . In order to start the algorithm, choose:

$$x^0 = (e; 1; 1)$$

$$\pi^0 = (0, \dots, 0) \text{ and } \pi_{m+1}^0 = \left(\frac{40\|c; M\|}{3} \right)$$

$$s^0 = \left(\frac{40\|c; M\|}{3} \right) (e, 1, 1) + (c; M; 0)$$

$$\theta^0 = \left(\frac{40\|c; M\|}{3} \right) .$$

Proposition 5.1 (x^0, π^0, s^0) is a $\beta = (\frac{3}{40})$ -approximate solution of $P(\theta^0)$ for the augmented problem $\tilde{P}$.

Proof: First notice that $x^0 > 0$ and that x^0 is feasible for $\tilde{P}$. Also, arithmetic manipulation shows that (π^0, s^0) is feasible for the dual $\tilde{D}$. Last of all, we must show that $\|(e, 1, 1) - (\frac{1}{\theta^0})X^0 s^0\| \leq \beta = \frac{3}{40}$. To see this, observe that

$$(e, 1, 1) - \frac{1}{\theta^0} X^0 s^0 = (e - e - \frac{c}{\theta^0}; 1 - 1 - \frac{M}{\theta^0}; 1 - 1 - 0) ,$$

and so

$$\left\| (e, 1, 1) - \frac{1}{\theta^0} X^0 s^0 \right\| = \frac{\|(c; M)\|}{\theta^0} = \frac{3}{40} ,$$

which completes the proof. ■

6 Exercises on Interior-Point Methods for LP

1. Verify that the formulas given in (11) indeed solve the equation system (10).
2. Consider the logarithmic barrier problem for the *dual* linear optimization problem (2):

$$\begin{aligned} D(\theta) : \quad & \max \quad q_\theta(\pi, s) := b^T \pi + \theta \sum_{j=1}^n \ln s_j \\ & \text{s.t.} \quad A^T \pi + s = c \\ & \quad \quad s > 0 . \end{aligned}$$

- a. Derive the optimality conditions for $D(\theta)$.
- b. Show that the optimality conditions for $D(\theta)$ can be rewritten as:

$$\begin{cases} Ax = b, \ x > 0 \\ A^T \pi + s = c \\ \frac{1}{\theta} X S e - e = 0 . \end{cases}$$

- c. Compare your result in (b.) to equation (4) in this chapter. What does this indicate about problems $P(\theta)$ and $D(\theta)$?
3. Verify that the formulas given in (20) indeed solve the equation system (19).
 4. Use Newton's method to solve the given logarithmic barrier problem:

$$\begin{aligned} P(\theta) : \quad & \text{minimize} \quad c^T x - \theta \sum_{j=1}^n \ln(x_j) \\ & \text{s.t.} \quad Ax = b \\ & \quad \quad x > 0 , \end{aligned}$$

for the fixed value $\theta = 6.0$, and

$$A = \begin{pmatrix} 1 & -1 & 3 & 1 \\ 1 & 2 & 1 & -1 \end{pmatrix} , \quad b = \begin{pmatrix} 6 \\ 3 \end{pmatrix} ,$$

and

$$c^T = \begin{pmatrix} 5 & 13.5 & 5.5 & -1 \end{pmatrix}.$$

Use the following starting point for your algorithm:

$$(x^0, \pi^0, s^0) = ((2, 1, 1, 2), (-1, 3), (3, 6.5, 5.5, 3)) .$$

Record the β values of your iterates, namely $\beta = \|\frac{1}{\theta} XSe - e\|$. What do you observe? Change the value of θ from $\theta = 6.0$ to nearby values of $\theta \in [5.0, 7.0]$ and re-run your algorithm. What do you observe now?

5. Implement Algorithm 4 to solve the following linear optimization problem:

LP:

$$\begin{aligned} & \text{minimize} && c^T x \\ & \text{s.t.} && Ax = b \\ & && x \geq 0, \end{aligned}$$

where

$$A = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & -1 & 0 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 100 \\ 50 \end{pmatrix},$$

and

$$c^T = \begin{pmatrix} -9 & -10 & 0 & 0 \end{pmatrix}.$$

In running your algorithm, you will need to specify the starting point (x^0, π^0, s^0) and the value of the fractional decrease parameter α . Use the following values and make runs of the algorithm for the four combinations of starting points and values of α given below:

- Starting Points:

$$\begin{aligned} & - (x^0, \pi^0, s^0) = ((1, 1, 1, 1), (0, 0), (1, 1, 1, 1)) \\ & - (x^0, \pi^0, s^0) = ((100, 100, 100, 100), (0, 0), (100, 100, 100, 100)) \end{aligned}$$

- Values of α :

$$\begin{aligned} & - \alpha = 1 - \frac{1}{10\sqrt{n}} \\ & - \alpha = \frac{1}{10} \end{aligned}$$

6. Suppose that $\bar{x} > 0$ is a feasible solution of the logarithmic barrier problem:

$$\begin{aligned} P(\theta) : \quad & \text{minimize} \quad c^T x - \theta \sum_{j=1}^n \ln(x_j) \\ & \text{s.t.} \quad Ax = b \\ & \quad \quad x > 0 , \end{aligned}$$

and that we wish to test if $\bar{x}$ is a β -approximate solution to $P(\theta)$ for a given value of θ . The conditions for $\bar{x}$ to be a β -approximate solution to $P(\theta)$ are that there exist values (π, s) for which:

$$\left\{ \begin{array}{l} A\bar{x} = b, \bar{x} > 0 \\ A^T \pi + s = c \\ \|\frac{1}{\theta} \bar{X} s - e\| \leq \beta . \end{array} \right. \quad (29)$$

Construct an analytic test for whether or not such (π, s) exist by solving an associated equality-constrained quadratic optimization problem (which has a closed form solution). HINT: Think of trying to minimize $\|\frac{1}{\theta} \bar{X} s - e\|$ over appropriate values of (π, s) .

7. Consider the logarithmic barrier problem:

$$\begin{aligned} P(\theta) : \quad & \min \quad f_\theta(x) := c^T x - \theta \sum_{j=1}^n \ln x_j \\ & \text{s.t.} \quad Ax = b \\ & \quad \quad x > 0 . \end{aligned}$$

Suppose $\bar{x}$ is a feasible point, i.e., $\bar{x} > 0$ and $A\bar{x} = b$. Let $(\Delta x, \pi')$ be the solution to the Newton equations (10) for $P(\theta)$ at $\bar{x}$. Let $\beta > 0$ be given. Show that $\|\bar{X}^{-1} \Delta x\| \leq \beta$ if and only if there exist $(\bar{\pi}, \bar{s})$ for which $(\bar{x}, \bar{\pi}, \bar{s})$ is a β -approximate solution of $P(\theta)$.

8. Consider the logarithmic barrier problem:

$$\begin{aligned} P(\theta) : \quad & \min \quad f_\theta(x) := c^T x - \theta \sum_{j=1}^n \ln x_j \\ & \text{s.t.} \quad Ax = b \\ & \quad \quad x > 0 . \end{aligned}$$

Suppose $\bar{x}$ is a feasible point, i.e., $\bar{x} > 0$ and $A\bar{x} = b$. Let $(\Delta x, \pi')$ be the solution to the Newton equations (10) for $P(\theta)$ at $\bar{x}$.

a. Show that Δx can be written as:

$$\Delta x = \bar{X}(I - \bar{X}A^T(A\bar{X}^2A^T)^{-1}A\bar{X}) \left(e - \frac{1}{\theta} \bar{X}\bar{s} \right) .$$

b. Show that the matrix $P := (I - \bar{X}A^T(A\bar{X}^2A^T)^{-1}A\bar{X})$ in **(a.)** is a projection matrix.

9. Let $\mathcal{F} := \{x \in \mathbb{R}^n : Ax = b, x \geq 0\}$, and suppose that $\bar{x} \in \mathcal{F}$ and $\bar{x} > 0$. Define the following ellipsoid:

$$\mathcal{E} := \{x \in \mathbb{R}^n : Ax = b, (x - \bar{x})^T \bar{X}^{-2} (x - \bar{x}) \leq 1\} .$$

Show that:

$$\mathcal{E} \subset \mathcal{F} .$$